

Versatile Audio-Visual Alarm

PETRE TZV PETROV

This circuit uses an NE555 timer IC, some LEDs, a couple of piezo buzzers and a few other components to produce audio-visual effects as per your requirement. The timer NE555 and its equivalents are widely used for all sorts of audio and visual indications, such as door alarms. But the sound produced by these circuits may not be always pleasant to hear, or the light produced may not be visually appealing. With this circuit you can get different audio-visual effects.

Here we use LEDs for visual indication and buzzers for audible alarms as they require relatively low current to operate. By simply connecting some resistors and capacitors to NE555 we can obtain some interesting visual and audible effects as described here.

Circuit and working

Fig. 1 shows the circuit of the versatile audio-visual alarm which is built

around timer NE555 (IC1), LEDs, buzzers and some resistors and capacitors. Resistors R1 and R2 and capacitor C1 determine the frequency of the LEDs' blinking. The frequency is selected usually within the range of 0.1Hz to 20Hz, depending on your requirement. Values of resistors R1 and R2 can be above 1-kilo-ohm. Capacitor C1's value can be between 1 μ F and 1000 μ F.

Timer NE555 drives two outputs, namely, Group1 and Group2. Group1 is built around resistors R4 and R6 along with LED1 through LED6. Group2 is built around resistors R7 and R8 along with LED7 through LED12.

Each of the groups can be configured to get different outputs. For example, in Group1 you can use only the LEDs (LED1 through LED3) connected to +12V, or only the LEDs (LED4 through LED6) connected to the ground, or both branches of these LEDs, or only piezo buzzer PZ1, or PZ1 with any combination of the LEDs,



or you can omit the entire Group1.

The components in Group2 can form the same combinations as the components in Group1. The difference between the Group1 and Group2 is the use of resistor R5 and capacitor C2. These two components give light-decay effect to the LEDs and a pleasant low-pitch sound to piezo buzzer in Group2. Value of resistor R5 can be between 75-ohm and 1-kilo-ohm and that of capacitor C2 between 47 μ F and 1000 μ F.

At point 1 (TP2) in the circuit you can see a rectangular wave signal. At point 2 you can see a triangular or trapezoidal-like signal. The signals at points 1 and 2 should go low, almost to zero, and should go high, almost to 12V supply voltage.

Power supply used is 12V, but it

Test Points

Test point	Details
TP0	0V
TP1	12V
TP2	Rectangular wave
TP3	Triangular wave

PARTS LIST

Semiconductors:

- IC1 - NE555 timer
- LED1-LED13 - 5mm LED
- D1 - 1N4148 signal diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3 - 10-kilo-ohm
- R2 - 1-mega-ohm
- R4, R6-R8 - 470-ohm
- R5 - 100-ohm
- R9 - 1-kilo-ohm

Capacitors:

- C1 - 4.7 μ F, 25V electrolytic
- C2 - 1000 μ F, 25V electrolytic
- C3 - 0.33 μ F ceramic disk
- C4 - 220 μ F, 25V electrolytic
- C5 - 0.1 μ F ceramic disk

Miscellaneous:

- CON1 - 2-pin terminal connector
- CON2 - 2-pin connector
- PZ1-PZ2 - Piezo buzzer with internal oscillators
- S1 - On/off switch
- 8-pin IC base

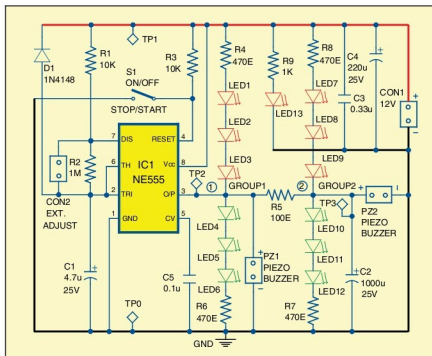


Fig. 1: The versatile audio-visual alarm circuit